CHRISTIAN OMAR PAYÁN TORRÓNTEGUI

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SUMMARY

Robotics & Digital Systems Engineering student focused on product design, advanced manufacturing, and robotics. Proven experience in 3D modeling, additive manufacturing, and embedded systems. Seeking to leverage mechanical design to bridge creative design with technical feasibility within the Specializing Master in Design Engineering and Innovation.

EDUCATION

B.S. Robotics and Digital Systems Graduating Expected December 2026
Tecnológico de Monterrey, Zapopan, Jalisco
School of Engineering and Sciences
Relevant coursework: Jewelry Concentration: Advanced 3D modeling and product design using Rhino 3D

TECHNICAL SKILLS

Programming & Languages: Python, C++, Spanish (Native), English (B2)
Robotics & Simulation: ROS 2, Gazebo, URDF
Design & Electronics: SOLIDWORKS, Fusion 360, Rhino 3D, Eagle
Tools & Methodologies: Git, Agile, Scrum, Advanced FDM 3D Printing, Linux (Ubuntu), Docker

PROFESSIONAL EXPERIENCE

Industrial Mounts & Robotic Support Interfaces, Zapopan, Jalisco: Mechanical & Product Designer Feb-Jun 2026

- Engineered mechanical interfaces for VR controllers (Meta Quest 3), developing a 3-DOF system that reduced fabrication costs by 95% vs commercial alternatives
- Designed and prototyped a custom mini PC enclosure for robust deployment in industrial environments, utilizing 3D-printed fiberglass-reinforced PETG for its ideal thermal and mechanical properties

ACADEMIC & TECHNICAL PROJECTS

Céruléen Ago-Dic 2025
Directed product design as part of the "Diseño, Producción y Comercialización de Joyería" concentration at Tec de Monterrey.

- Directed 3D modeling (Rhino 3D) for 2 collections; exhibited "The Perverse Beauty" at NYCJW 2025 ("Echoes of Nature")
- Integrated digital fabrication with artisanal processes, optimizing product life cycle and reducing production time by 20%

Open-Source Gastrointestinal Medical Phantom Ago-Dic 2025 – Feb-Jun 2026
Designed an open-source medical phantom for endoscopy training.

- Engineered phantom via advanced FDM 3D printing (TPU/PETG), reducing prototyping costs by 80% vs commercial alternatives
- Managed development using Agile methodologies to iterate rapidly on anatomical accuracy and material constraints

Port Automation & Drone Sim *(github.com/opyntorr/port_bot_sim_ws, .../demo_tello_sim)* Ago-Dic 2025
Developed a port simulation environment utilizing ROS 2 and Gazebo in partnership with OMRON.

- Engineered URDF models and C++/Python ROS 2 nodes for kinematic control and reliable flight telemetry
- Designed & 3D-printed custom hardware: drone mirror/lens mounts, and an AMR bracket (LiDAR, ArUco, Jetson)

DigiMix *(github.com/FrediRomo/DigiMix)* Ago-Dic 2024
Designed an embedded system integrating custom hardware and digital interfaces with Intel.

- Designed custom PCB and 3D-printed casing; engineered hardware using dual-core STM32 and ESP32 (WebSockets)
- Applied Scrum to manage lifecycle, integrating hardware and software architectures with zero logic faults